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Project Title:	Cytoplasmic/Golgi interactions in the nanofabrication of biomineral composites - a genetic approach using coccolithophores		
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Task Description:

1. The first task is to identify macromolecular elements associated with the cytoplasm and Golgi which regulate the nucleation and growth of coccolith crystals (calcite). Our approach is to generate and isolate coccolithophore mutants using RNAi which do not express a particular macromolecule - one with a putative mineralizing function - and determine how its absence affects the structure of the biocomposite in vivo.
2. The second task is to explore the effects of nucleating agents and crystal shaping elements identified in the first task on calcite nucleation and growth in vitro - both in bulk solution and in synthetic liposomes.

3. The third task is a study of coccolithophore phase transition between calcifying and noncalcifying morphotypes. The overall goal of this project is to determine how CaCO3 formation is regulated in coccolithophores, calcifying phytoplankton responsible for the majority of the CaCO3 formed in the open ocean since the early Cretaceous. Coccolithophores impact the ocean carbon cycle and may regulate the Quaternary pCO2 flux through transitions between calcifying and noncalcifying morphotypes. An understanding of the cellular processes, genetic elements, and environmental cues which regulate the expression of calcifying versus noncalcifying phases will be essential for predicting future climate change. Regulation of CaCO3 crystal size and shape by coccolithophores is relevant to nanofabrication, as biological processes can provide inspiration for development of novel and useful materials such as molecular sieves and catalysts.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Biomineralization in coccolithophores is a phenomenon of global import. The appearance of coccolithophores and foraminifera — calcareous nannoplankton — in the late Triassic marked the onset of calcium carbonate sedimentation in the deep ocean, a process thought to stabilizes the carbon cycle today. Coccolithophores are the major produces of pelagic CaCO3 and impact the atmosphere by generating CO2 during calcification. $Ca2+ + 2HCO3– = CaCO3 + CO2 + H2O$. The global carbon cycle is a major societal concern because the extraction and burning of fossil fuels is transferring carbon from the slowing exchanging lithosphere into the atmosphere as CO2, a greenhouse. Theoretically global mean temperature will rise as atmospheric pCO2 increases, a theory consistent with ice core analyses showing a tight correlation between atmospheric pCO2 and sea surface temperature over the past 400,000 years. The ocean is the major reservoir for absorbing atmospheric CO2 changes; the gas is partitioned between atmosphere and ocean through planktonic photosynthesis and CaCO3 formation and by ocean circulation with the exchange of carbon between deep and surface waters. An understanding of how coccolithophore calcification impacts the natural Quaternary pCO2 flux and is impacted by anthropogenic CO2 emissions will be essential for prediction of future climate change.

Task Progress:

We have analyzed the acidic polysaccharides of two new coccolithophore species. Both species — Cruciplacolithus neohelis and Gephyrocapsa oceanica — synthesize a sulfated galacturonomannan which surrounds the surface of the coccolith crystals. Previously we had shown that this galacturonomannan (PS3 in Pleurochrysis carterae and CP in Emiliania huxleyi) is part of the machinery which induces growth and development of the CaCO3 (calcite) protocrystals into the elaborate and intricately shaped crystals of the mature coccoliths (the calcified scales of the coccolithophores). Pleurochrysis mutants which do not express PS3 form only small protocrystals with simple shapes (rhombs) characteristic of inorganically formed calcite. With this new data from distantly related species, we are confident that the sulfated galacturonomannans are central elements in the nanofabrication of CaCO3 crystals in all placolith-bearing coccolithophores.
However the same generalization cannot be applied to PS2, a highly acidic tartrate- and glyoxylate-containing polysaccharide which has an enormous capacity for sequestering calcium ions. PS2 is the most abundant mineral-associated polysaccharide in Pleurochrysis where it functions as a calcium buffer, maintaining the calcium ion concentration within the mineralizing vesicle at an adequate level to support the nucleation and growth of CaCO3 crystals. Pleurochrysis mutants which do not synthesize PS2 form very few crystals (less than 5% of the number formed by wild-type cells). However PS2 was not found in Emiliania, Cruciplacolithus, or Gephyrocapsa. This implies that species other than members of the family Pleurochrysidaceae must mineralize by a PS2-independent mechanism, a rather surprising result given the structural similarity of the coccoliths of all four species. PS2-independence outside the Pleurochrysidaceae indicates that other coccolithophore families are able to efficiently take up calcium directly from the environment for calcite formation and do not require an internally sequestered calcium ion reservoir.

The PS2-dependent and PS2-independent CaCO3 formation pathways in coccolithophores may differentially impact the ocean/atmosphere climate system if one pathway is more robust than the other. When coccolithophores fail to mineralize — that is when they shift from mineralizing to nonmineralizing morphotypes — their environment (the ocean) becomes more basic and more CO2 is drawn down from the atmosphere into the ocean. We and others have noted that the mineralizing morphotype of Pleurochrysis (PS2-dependent) is much more stable than the calcifying phase Emiliania (PS2-independent) both in culture and in the ocean. Transition to a nonmineralizing morphotype in diploid Pleurochrysis is rare, and when it does occur, it is often accompanied by loss of PS2 synthesis. On the other hand, Emiliania commonly shifts into stable nonmineralizing morphotypes. The genetic basis of the Emiliania transitions has not been determined, but it is probable that some element of the PS2-independent pathway is perturbed. It appears that the PS2-dependent mechanism may be a more reliable calcifying system; however, morphotype stabilities in other PS2-independent species have not yet been explored.

We are developing reagents to identify genes required for morphotype transition and CaCO3 formation in coccolithophores. In order to use RNA interference (RNAi) technology to suppress — and hence identify — these genes, a transformation vector specific for coccolithophore species is being developed; a vector capable of expressing double stranded RNA from inserts obtained from the coccolithophore cDNA. Thus far we have isolated three strong promoters (and their terminators) which drive expression of three members of the multiple gene family encoding fucoxanthin chlorophyll a/c-binding proteins (fcp genes) and have identified two antibiotics — genetin and zeocin/phleomycin — to which the coccolithophores are sensitive and for which resistance genes are available. Now we are constructing a vector with tandem cassettes to allow the simultaneous expression of an antibiotic resistance gene and a reporter gene or an RNAi construct. The cassette for the RNAi construct will consist of opposing fcp promoters driving the transcription of both the sense and antisense chains of cDNA derived from the reverse transcription of the coccolithophore mRNA. Individual cells will harbor just one cDNA insert, and cell population will be a complete coccolithophore RNAi library, covering the entire transcriptome.

Bibliography Type:	Description: (Last Updated: 09/11/2019)
Articles in Peer-reviewed Journals	Henriksen K, Stipp SLS, Young JR, Marsh M. "Biological control on calcite crystallization: AFM investigation of coccolith polysaccharide function." American Mineralogist. 2004 Nov;89(11-12):1709-16. https://doi.org/10.2138/am-2004-11-1217 , Nov-2004
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